Abu Dhabi City Campus Daily/Weekly Planner

مدارس الإمارات الوطنية
Emirates National Schools

Week 8: 19–23 October 2026

Subject: Programming **Unit Title:** Python Fundamentals: Input, Conversion, Operators, Strings, Boolean Logic & Conditionals **Grade:** 12

Lessons	Learning targets	Assessment tools and strategies Formative / Summative / Differentiation	Resources and links for Parents
Lesson 1 **User input and data conversion**	Learning Target: Collect values with input() and convert strings to numeric data types when calculations are needed. Focus: input(), int(), float(), and why input() returns text.	**Formative:** Predict input type; conversion matching; debug an age-entry snippet; exit ticket. **Summative:** Short input-and-conversion code trace feeding into the end-of-week challenge. **Support:** Modelled syntax cards, annotated examples and a conversion word bank. **Core:** Write prompts and appropriate conversions in simple code. **Extension:** Explain conversion failures and suggest safe input checking.	Guided Python examples; teacher-approved IDE or paper tracing; data-type cards; LMS practice; optional home discussion about input errors.
Lesson 2 **Mathematical operators and calculations**	Learning Target: Use arithmetic operators with converted input values to calculate accurate results. Focus: +, -, *, /, //, %, ** and order of operations in authentic calculations.	**Formative:** Operator sort, prediction grid, pair code trace and mini-quiz. **Summative:** A small input–process–output price calculator completed independently. **Support:** Worked equations, visual operator reference, partially completed code. **Core:** Calculate and display results using int() or float() as appropriate. **Extension:** Compare /, // and % and justify the most appropriate operator.	Operator reference sheet; worked cost scenarios; school-approved coding environment; calculator for checking; LMS examples.
Lesson 3 **Strings and Boolean logic**	Learning Target: Manipulate string data and build Boolean expressions that evaluate to True or False. Focus: text concatenation, f-strings, len(), .strip(), comparisons, and/or/not.	**Formative:** String-output predictions, Boolean truth-table sort, short collaborative debugging. **Summative:** A mini scenario requiring formatted text output and a Boolean check. **Support:** Sentence frames, string examples, colour-coded true/false cards. **Core:** Join strings and combine comparisons using and/or/not. **Extension:** Compare alternative Boolean expressions and explain edge cases.	String and Boolean reference cards; truth tables; fictional club-registration data; digital/paper code traces; home discussion about clear program messages.
Lesson 4 **Conditionals: making decisions**	Learning Target: Use input, conversion, operators, strings and Boolean conditions to make appropriate program decisions. Focus: if / elif / else, indentation, comparisons and integrated mini-program.	**Formative:** Branch-flowchart annotation; predict output; teacher feedback on program drafts. **Summative:** Week 8 mini-program task: a fictional school activity booking calculator with eligibility or discount rules (teacher-set assessment). **Support:** Structured if/else template, condition prompts and worked model. **Core:** Write and test a working decision-making program. **Extension:** Handle invalid inputs and justify multiple test cases.	If/else flowcharts; teacher-approved Python IDE; printable differentiated worksheets; sample test data; LMS submission; no real personal data.
Homework	Review input(), int(), float(), arithmetic operators, strings, True/False, and if/elif/else. Trace one short program, predict its output and explain the condition used.	**Formative:** Short retrieval quiz, annotated code trace, and self-check against Must / Should / Could.	LMS concept revision; Python syntax sheet; optional family discussion about real-life decisions computers make using input.
Teacher's notes / Reminders	Sequence supplied by the teacher for Week 8. Start with simple executable models; build toward one integrated conditional program. Print worksheets for non-device lessons. Use fictional names only; never collect real personal information.		

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Technology Integration	Approved Python editor/IDE or offline code tracing; LMS assignments; short demonstrations; debugging checks; truth-table activities; flowcharts; digital exit tickets.
Cross-curricular links	Mathematics: calculations, operators, inequalities, logical expressions and order of operations. English: precise user prompts, formatted outputs, clear explanations and debugging communication. Business: accurate transaction calculations.
External Assessment Integration	Interpret Python code, identify data types, calculate expressions, predict outputs, correct syntax/logic errors, construct a Boolean condition and implement an if/elif/else decision from a scenario.
Wellbeing / ATL Skills	Computational thinking; resilience in debugging; collaboration; communication; responsible use of technology; self-management; problem solving; precision and reflection.
Standards	Teacher-provided Week 8 Programming concepts: User Input, Data Conversion, Mathematical Operators, Strings, Boolean Logic, Conditionals. Skills: trace, write, test and debug short Python programs.
Inquiry / TOK Question	How can the way a program collects, converts and tests information change the fairness and accuracy of its decisions?
Success Criteria (Measurable)	Week 8 learning target: I can write, trace and debug simple Python programs that take user input, convert values, calculate results, work with strings, test Boolean conditions and make decisions. Lesson 1 — MUST: state that input() returns text. SHOULD: select int() or float() for a task. COULD: diagnose a failed conversion. Lesson 2 — MUST: identify common operators. SHOULD: calculate accurate expressions using converted input. COULD: justify //, % or ** in a scenario. Lesson 3 — MUST: identify string and Boolean outputs. SHOULD: format text and combine comparisons with and/or/not. COULD: explain equivalent expressions or edge cases. Lesson 4 — MUST: identify an if condition and its branches. SHOULD: write a working if/elif/else program. COULD: validate assumptions and justify test cases.
National Identity Domain	UAE values: responsibility, innovation, efficiency and trustworthy technology. Area: digital innovation and responsible decision-making. Implementation: code a fictional UAE school service, calculate transparent results and test rules fairly without collecting real learner data.